SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA0501**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

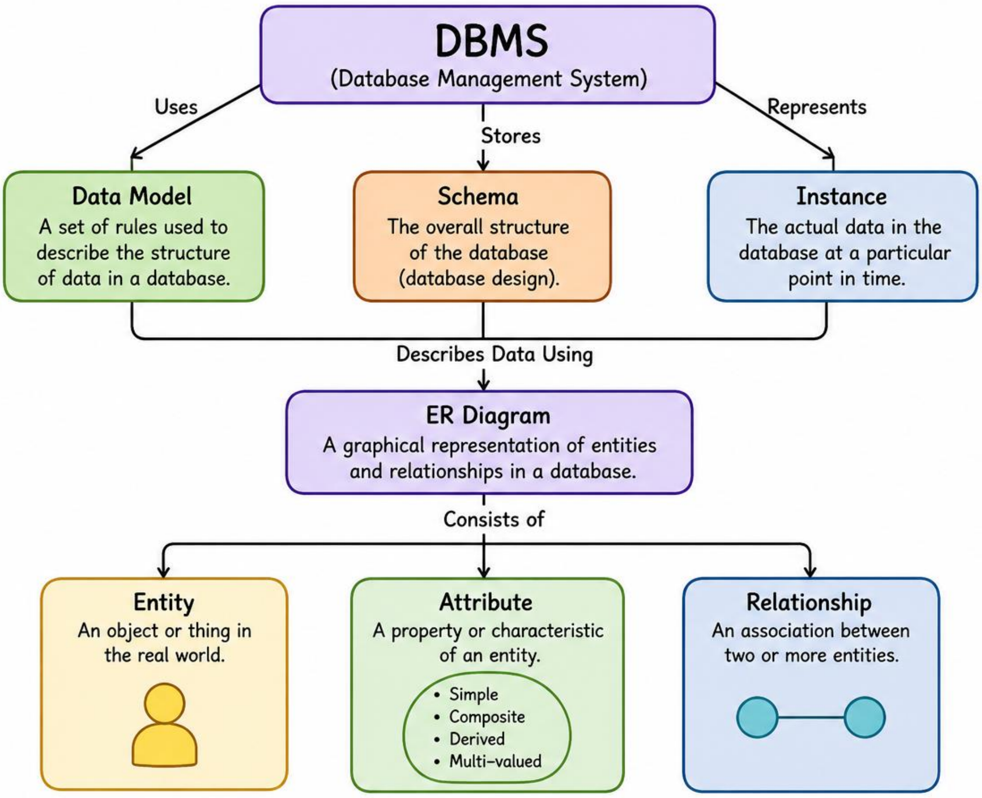
Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 35%

Dr

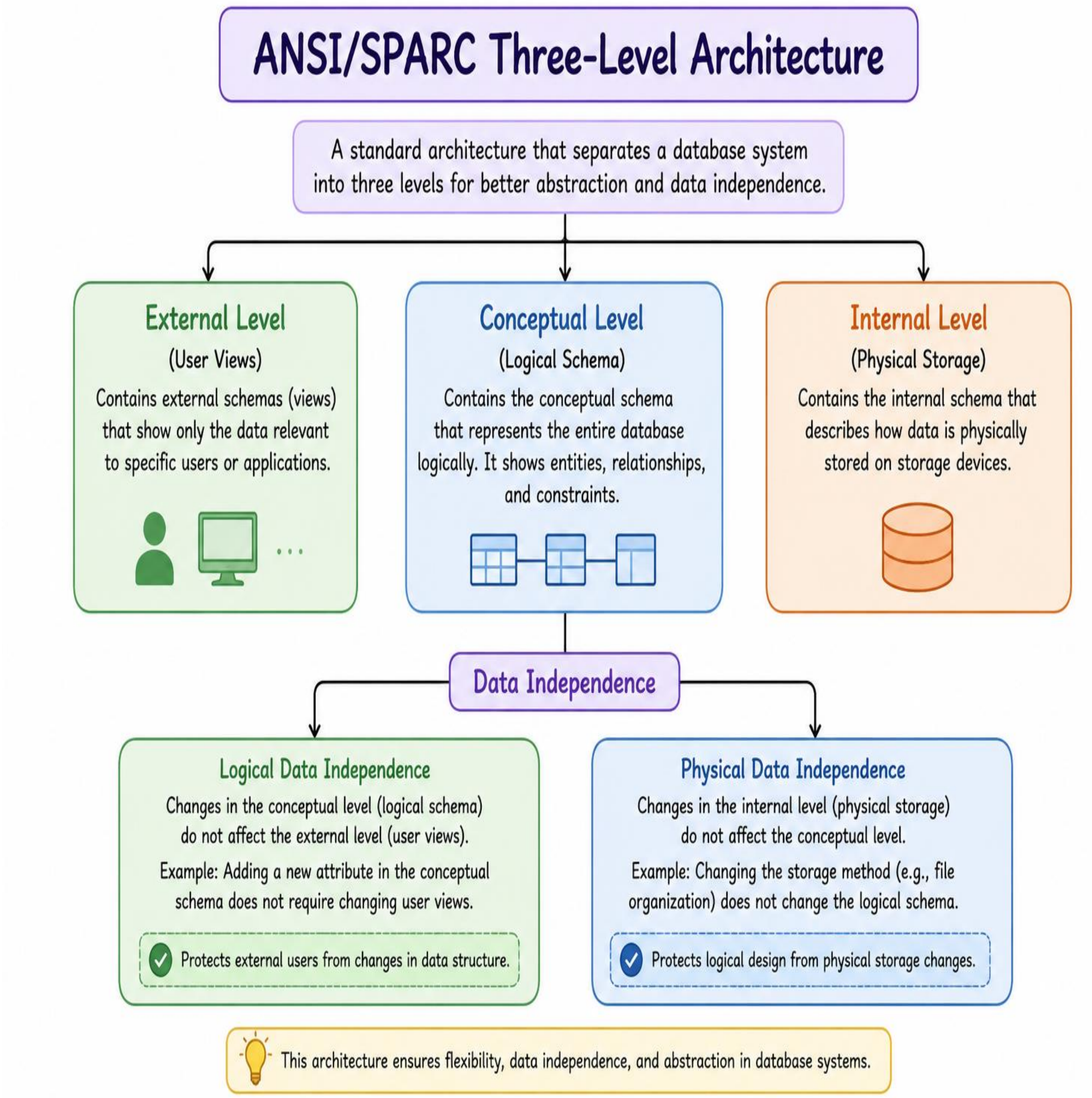
QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	**5**
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	**5**
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	**5**
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	**5**
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	**5**
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	**5**

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	**5**
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	**5**
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	**5**
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	**5**

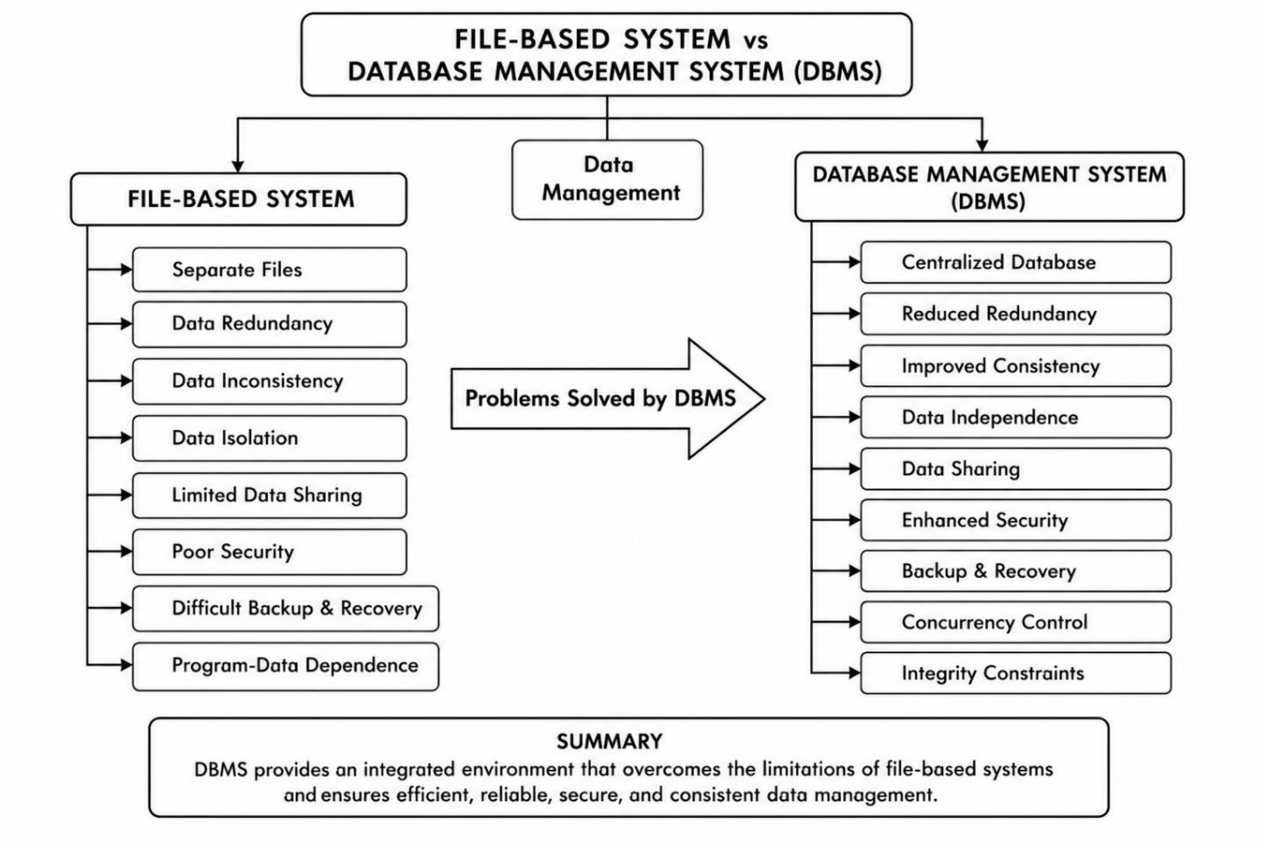
1. DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship



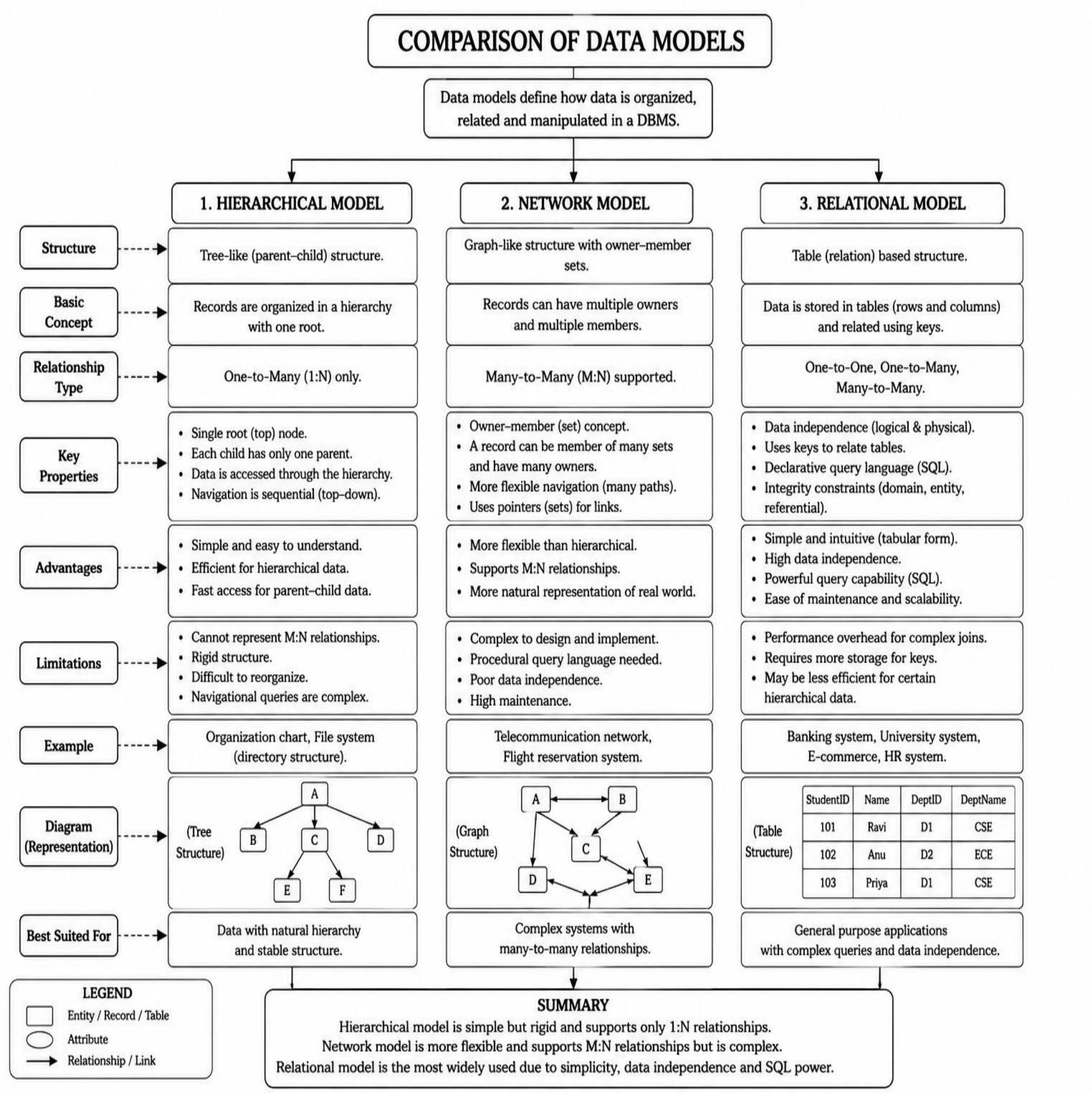
2.Three level ANSI/SPARC Architecture



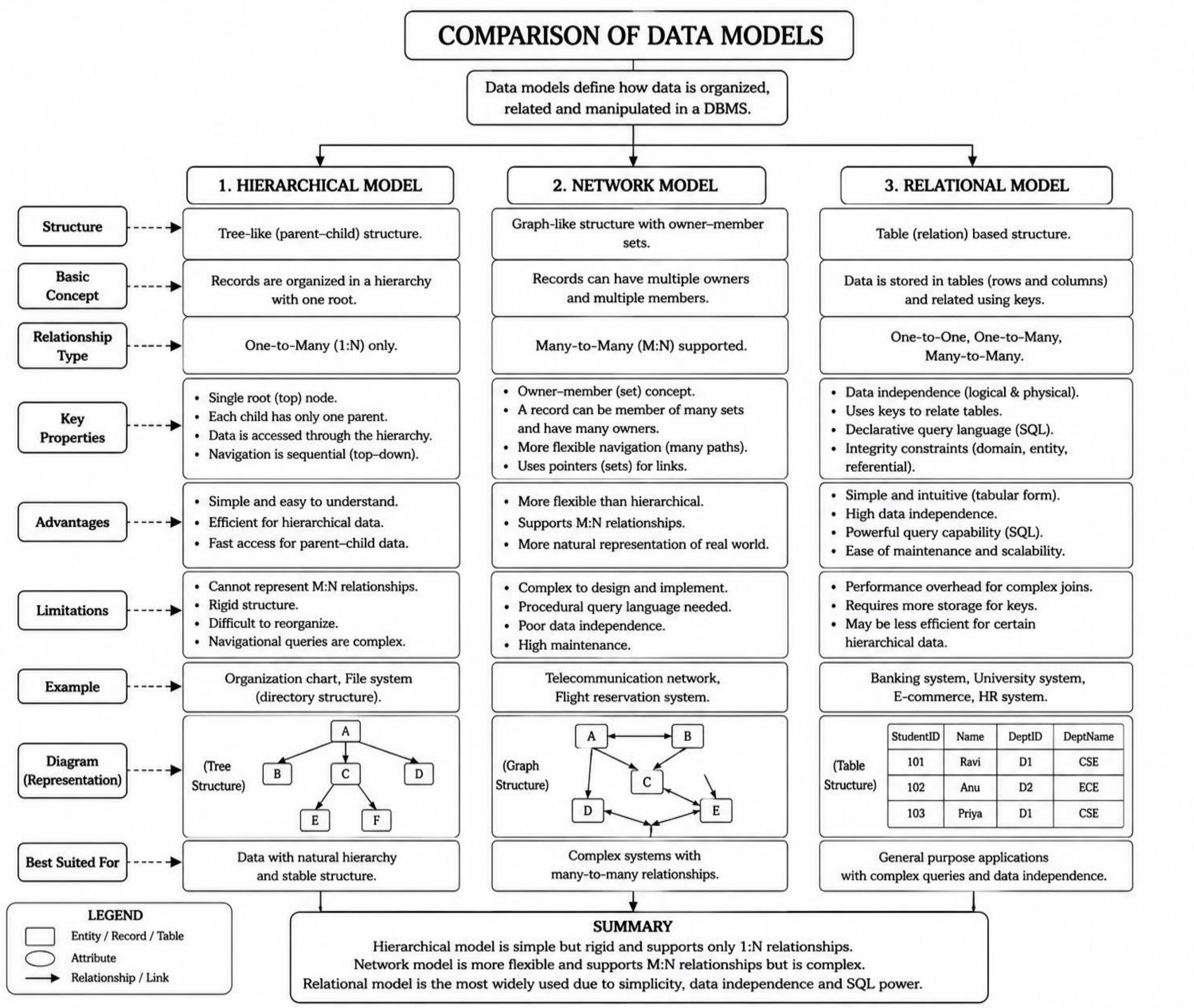
3.File Based System vs Database Mnagement Sytem



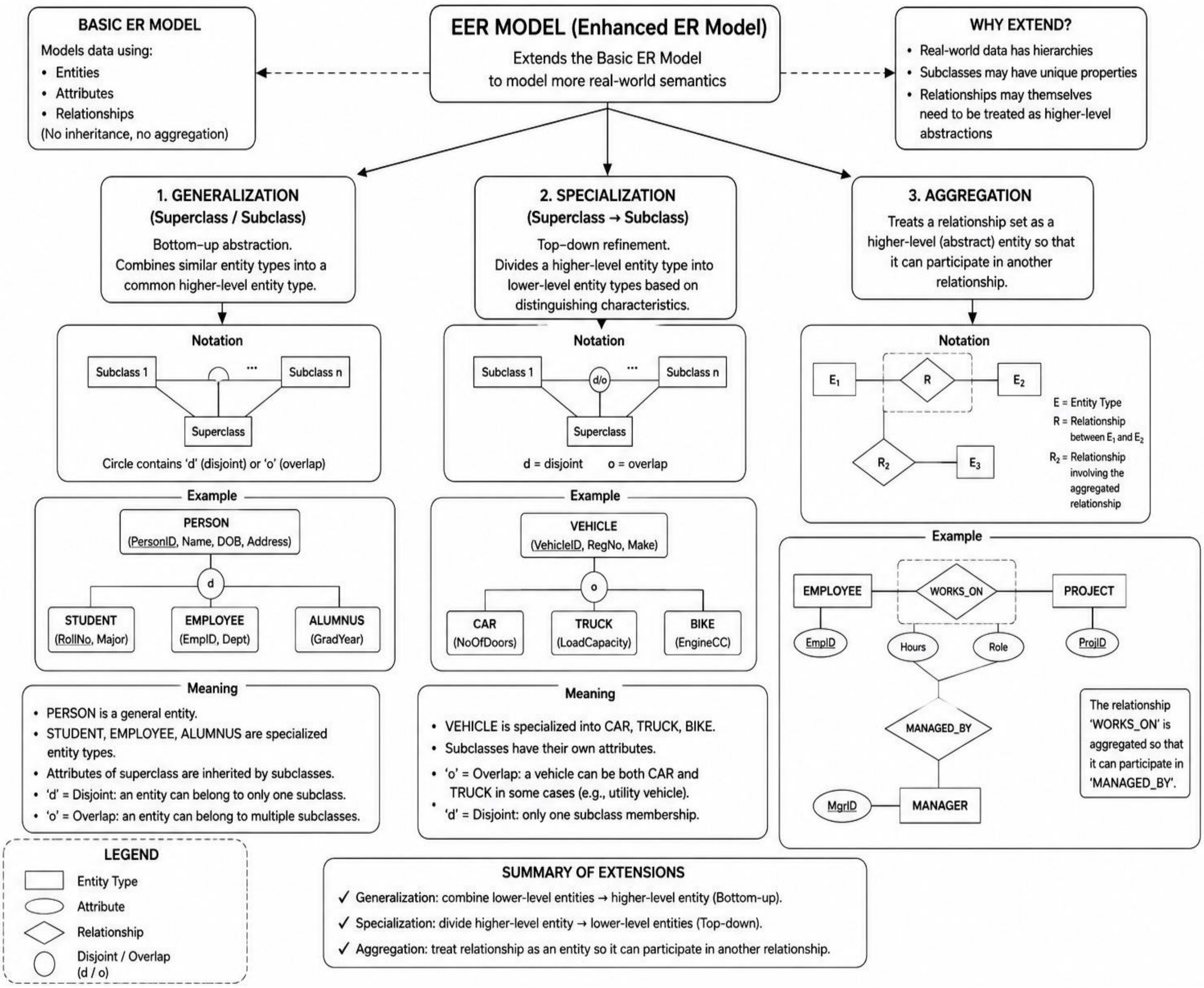
4.ER Diagram Components



5.comparing Hierarchical, Network, and Relational data models .

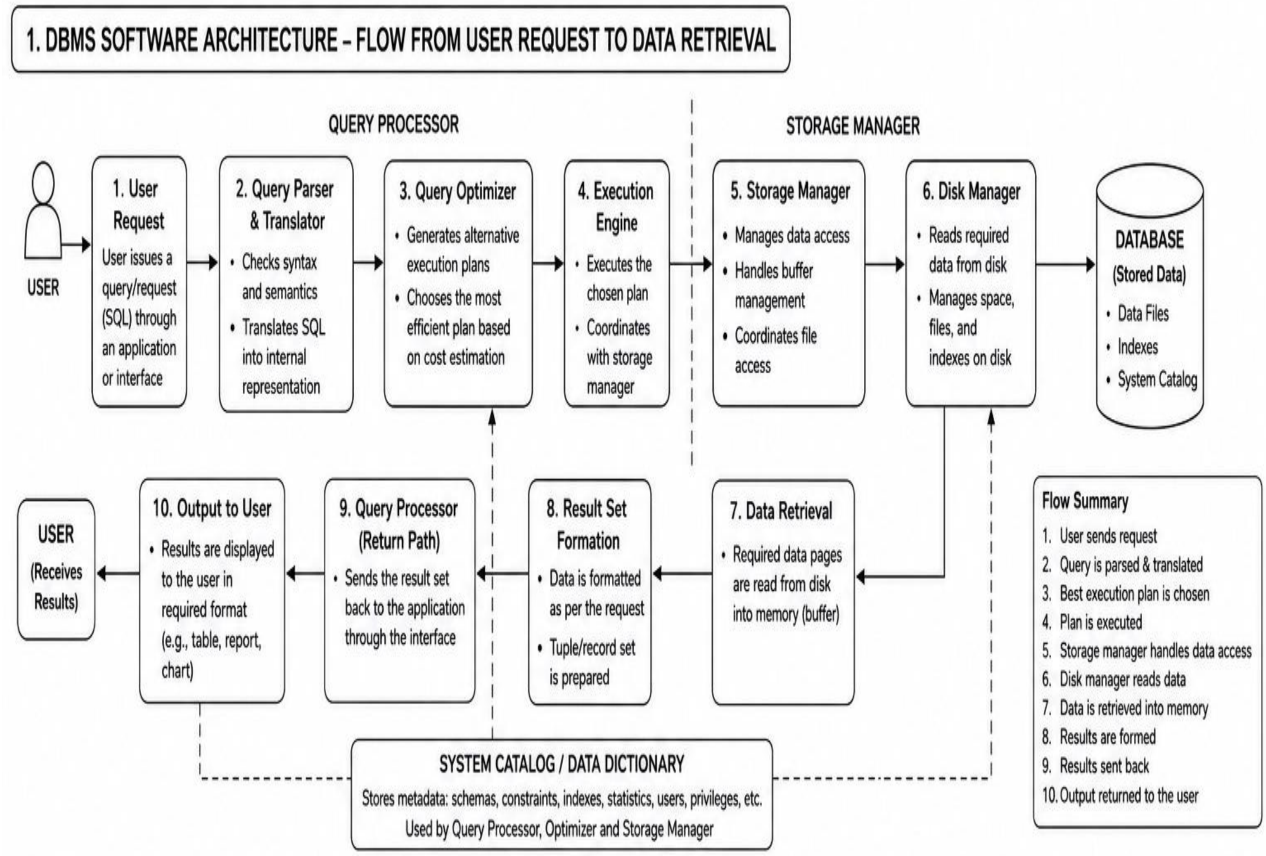


6.EER Model

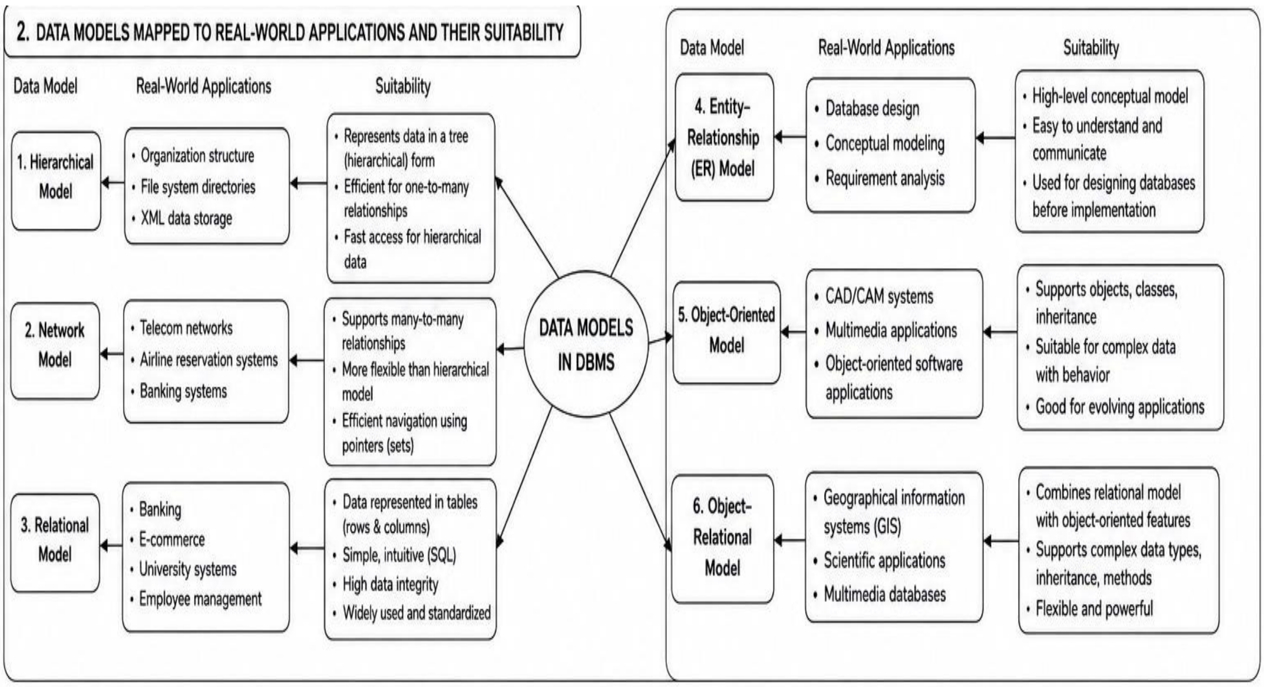


BASIC ER MODEL
Models data using:
• Entities
• Attributes
• Relationships
(No inheritance, no aggregation)
EER MODEL (Enhanced ER Model)
Extends the Basic ER Model to model more real-world semantics
WHY EXTEND?
• Real-world data has hierarchies
• Subclasses may have unique properties
• Relationships may themselves need to be treated as higher-level abstractions
1. GENERALIZATION (Superclass / Subclass)
Bottom-up abstraction. Combines similar entity types into a common higher-level entity type.
2. SPECIALIZATION (Superclass → Subclass)
Top-down refinement. Divides a higher-level entity type into lower-level entity types based on distinguishing characteristics.
3. AGGREGATION
Treats a relationship set as a higher-level (abstract) entity so that it can participate in another relationship.
Notation
Subclass 1
Subclass n
Superclass
Circle contains 'd' (disjoint) or 'o' (overlap)
Notation
Subclass 1
d/o
Subclass n
Superclass
d = disjoint o = overlap
Notation
E1
R
E2
R2
E3
E = Entity Type
R = Relationship between E1 and E2
R2 = Relationship involving the aggregated relationship
Example
PERSON
(PersonID, Name, DOB, Address)
d
STUDENT (RollNo, Major)
EMPLOYEE (EmpID, Dept)
ALUMNUS (GradYear)
Example
VEHICLE
(VehicleID, RegNo, Make)
o
CAR (NoOfDoors)
TRUCK (LoadCapacity)
BIKE (EngineCC)
Example
EMPLOYEE
WORKS_ON
PROJECT
EmpID
Hours
Role
ProjID
MANAGED_BY
MgrID
MANAGER
The relationship 'WORKS_ON' is aggregated so that it can participate in 'MANAGED_BY'.
Meaning
• PERSON is a general entity.
• STUDENT, EMPLOYEE, ALUMNUS are specialized entity types.
• Attributes of superclass are inherited by subclasses.
• 'd' = Disjoint: an entity can belong to only one subclass.
• 'o' = Overlap: an entity can belong to multiple subclasses.
Meaning
• VEHICLE is specialized into CAR, TRUCK, BIKE.
• Subclasses have their own attributes.
• 'o' = Overlap: a vehicle can be both CAR and TRUCK in some cases (e.g., utility vehicle).
• 'd' = Disjoint: only one subclass membership.
LEGEND
Entity Type
Attribute
Relationship
Disjoint / Overlap (d / o)
SUMMARY OF EXTENSIONS
✓ Generalization: combine lower-level entities → higher-level entity (Bottom-up).
✓ Specialization: divide higher-level entity → lower-level entities (Top-down).
✓ Aggregation: treat relationship as an entity so it can participate in another relationship.

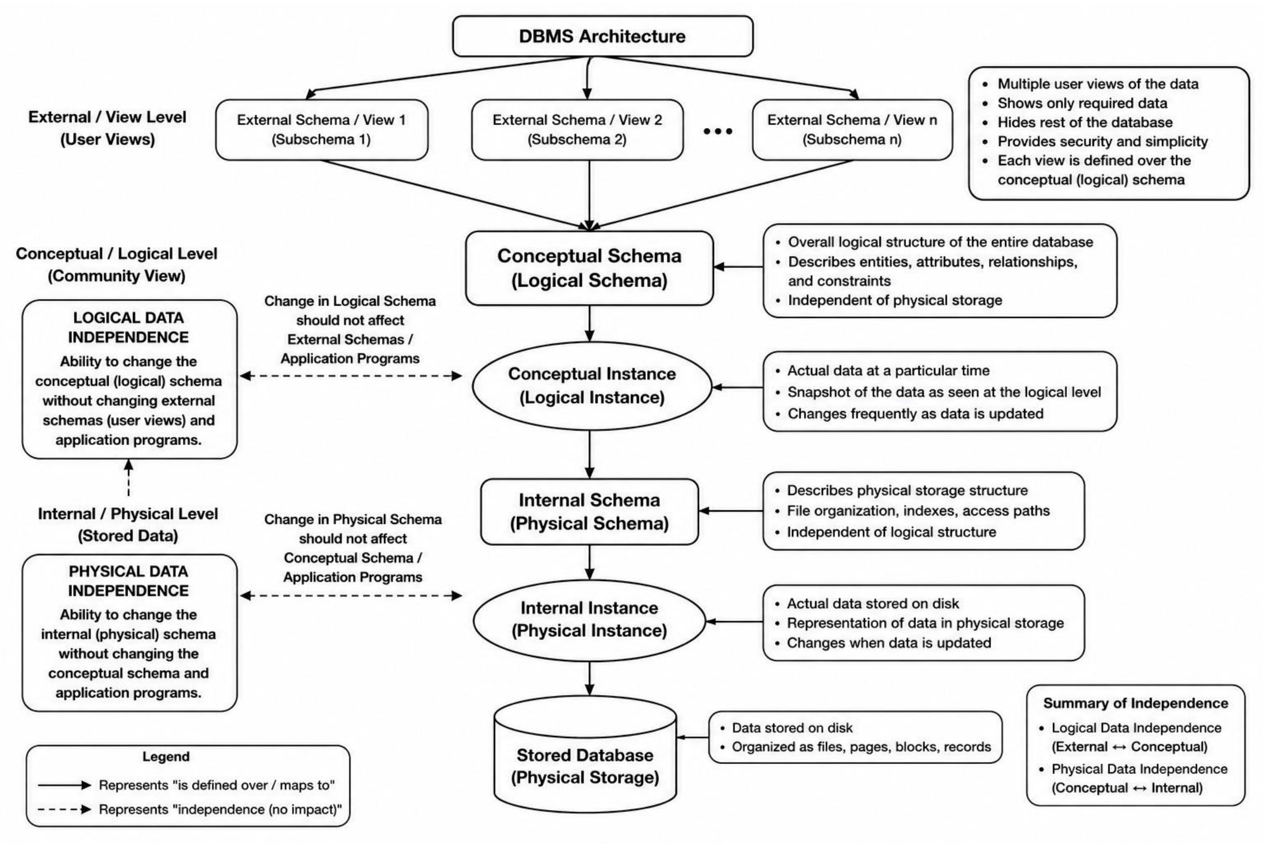
7.DBMS SOFTWARE ARCHITECTURE



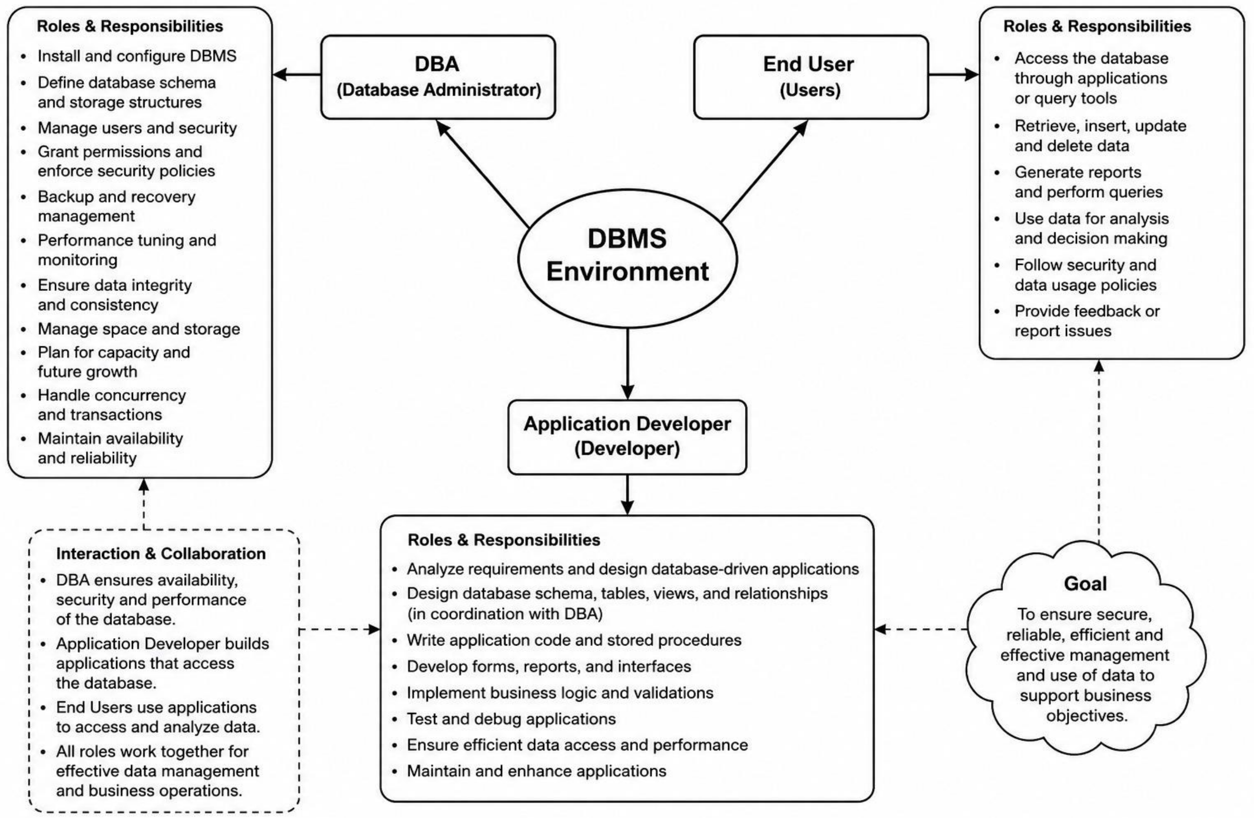
8 . DATA MODELS IN DBMS



9.DBMS Architecture



10.DBMS ENVIRONMENT



Code of Conduct:

I **NS AVESH HUSSAIN(192525241)***Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
NS AVESH HUSSAIN

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand